

Vision Transformer

Lukas Galke Poech, AI 508, Spring 2026

Task of the day

Lecture 1: Introduction

Lecture 2: Vision Transformer

Lecture 3: You Only Look Once

Lecture 4: Diffusion

Lecture 5: Segment Anything

Lecture 6: Vision Language Models

Keywords

- Image classification
- Residual Networks (ResNet)
- EfficientNet
- Vision Transformer (ViT)
- Swin Transformer
- Representation learning

Image Classification

What is image classification?

Given an image predict its class (or category).

In math: Estimate $p(y|x)$, where y is a class label and x is an image.



The ImageNet dataset

A major database of images and class labels

Around 14.2M images, around 22k classes

Facilitated the breakthrough of the convolutional neural network AlexNet in the ImageNet challenge 2012 – often cited as the beginning of the deep learning revolution.

ballplayer 50.61%



anemone_fish 97.65%



African_elephant 52.34%



forklift 98.88%



ice_cream 96.31%



lemon 94.01%



combination_lock 35.00%



ice_bear 99.48%



Training a model for image classification

1. Choose an architecture
2. Output dimension of last layer = #classes
3. Apply softmax, and minimize cross-entropy loss of outputs against true class labels:

$$H(P, Q) = - \sum P(y|x) \cdot \log Q(y|x)$$

where $P(y|x)$ is 1 when it's the true class and 0 otherwise, and $Q(y|x)$ are the (softmax'ed) estimated probabilities from the model.

$$\text{softmax}(y)_i = \frac{e^{y_i}}{\sum_j^K e^{y_j}}$$

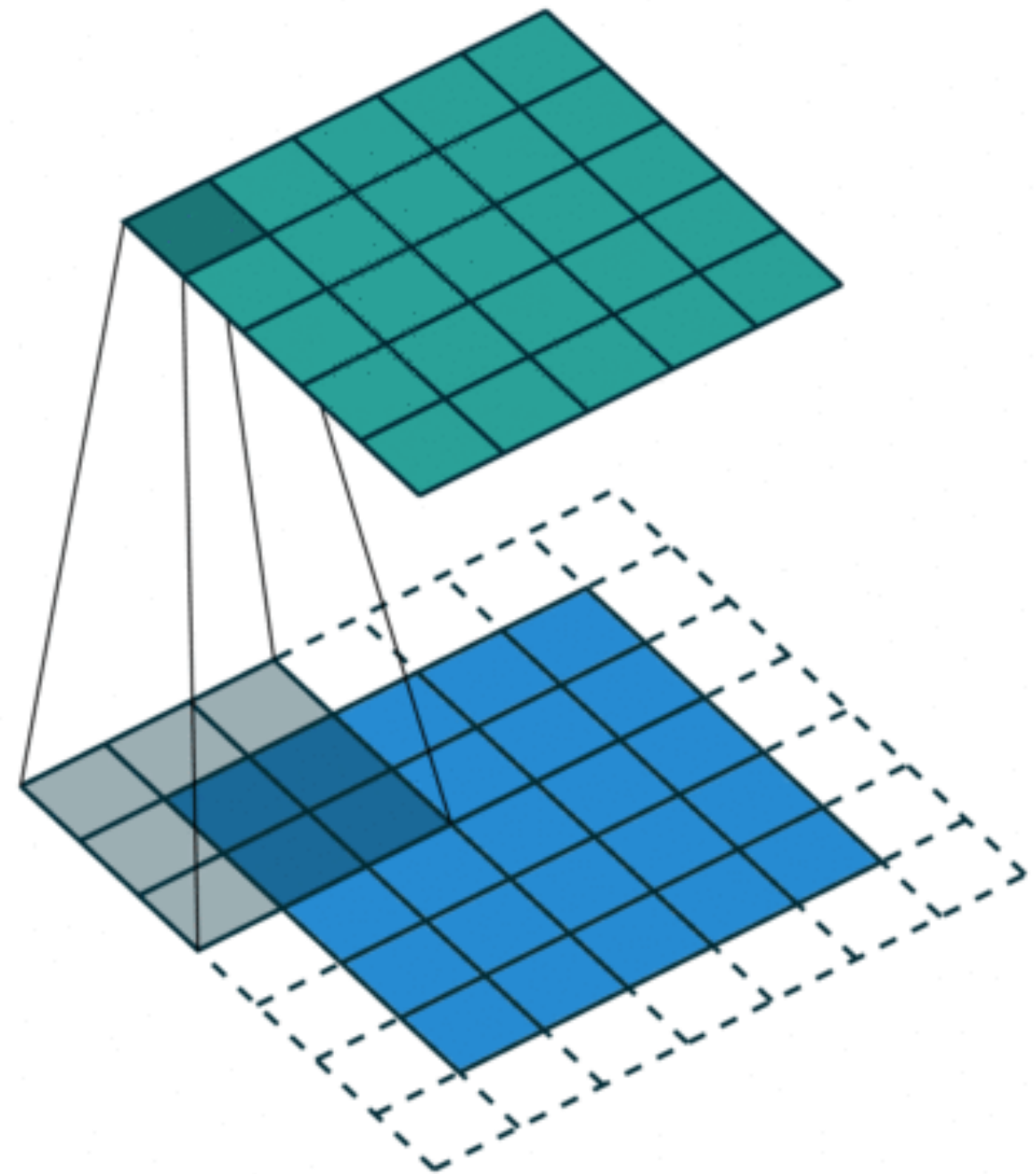
(makes sure probs are all positive and sum up to one)

Convolutional Networks

Quick recap of convolutional networks

Key components:

- Convolution: Learned filter to sweep over the image
- Pooling: Aggregation operation to reduce image dimensions

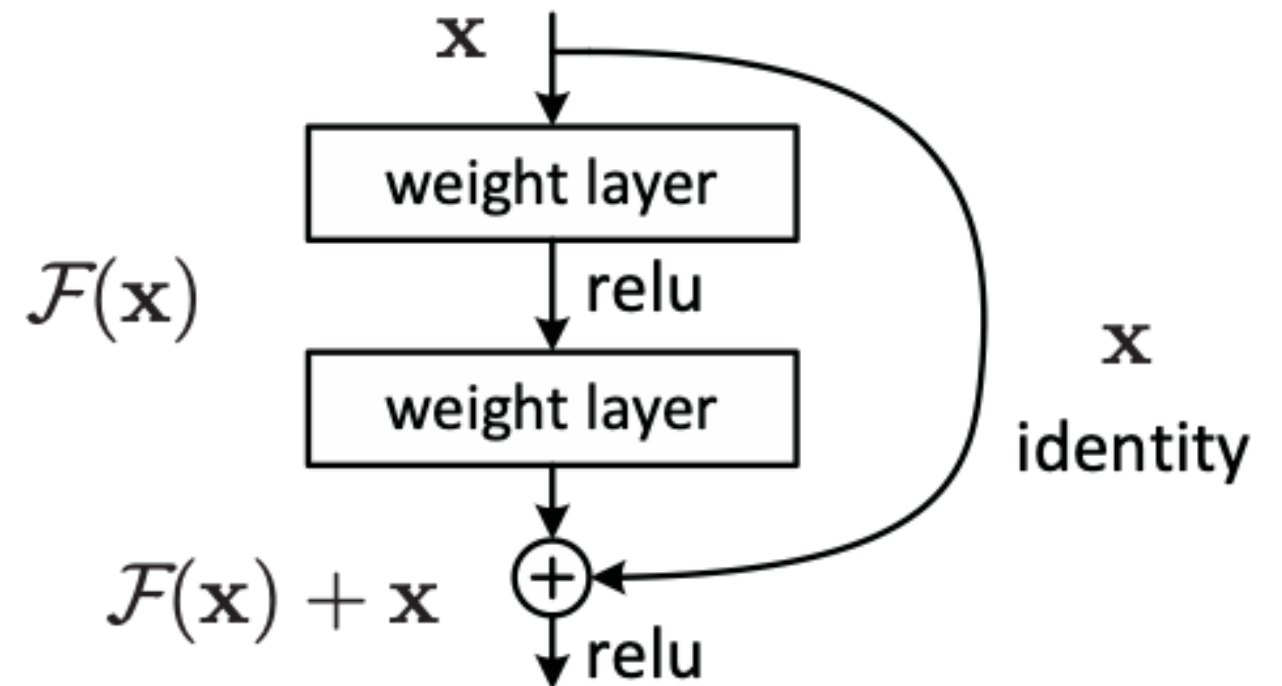


Residual Networks

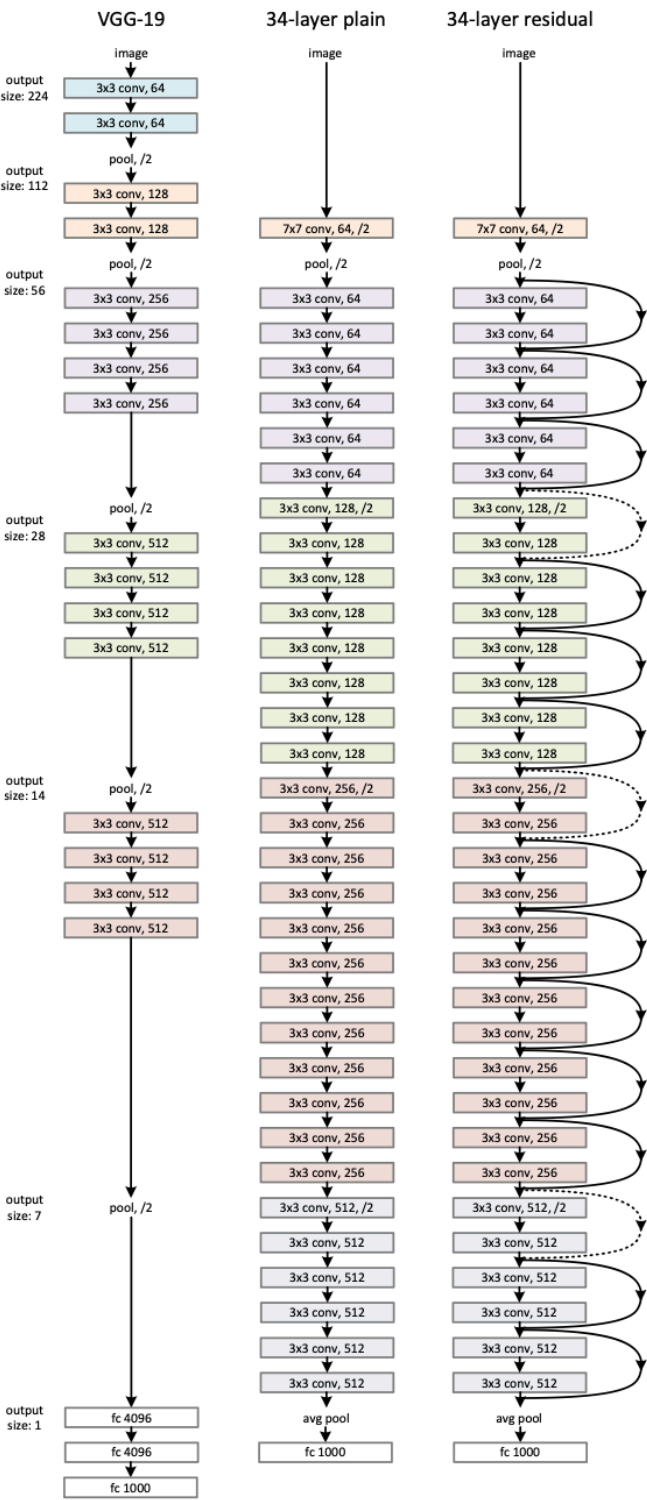
Residual connections

Residual connections are the core idea of residual networks.

Allows training of deeper networks due to better gradient flow.



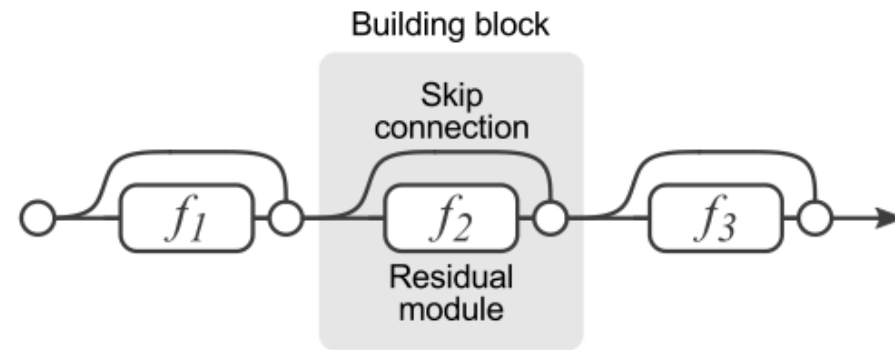
ResNet architecture



ResNet was the state of the art in image classification for a long time.

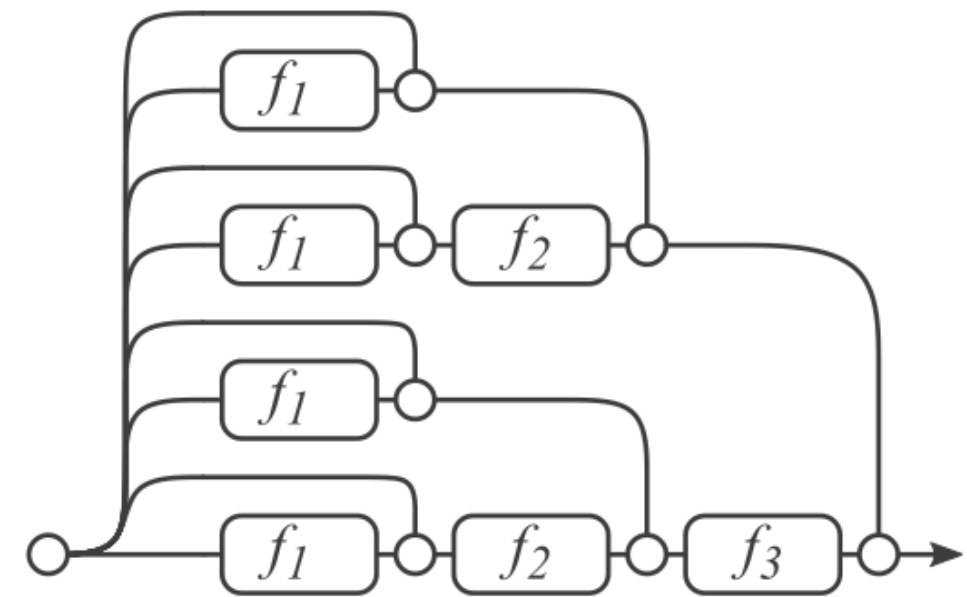
method	top-1 err.	top-5 err.
VGG [41] (ILSVRC'14)	-	8.43 [†]
GoogLeNet [44] (ILSVRC'14)	-	7.89
VGG [41] (v5)	24.4	7.1
PReLU-net [13]	21.59	5.71
BN-inception [16]	21.99	5.81
ResNet-34 B	21.84	5.71
ResNet-34 C	21.53	5.60
ResNet-50	20.74	5.25
ResNet-101	19.87	4.60
ResNet-152	19.38	4.49

Residual Networks Behave Like Ensembles of Relatively Shallow Networks¹



(a) Conventional 3-block residual network

=

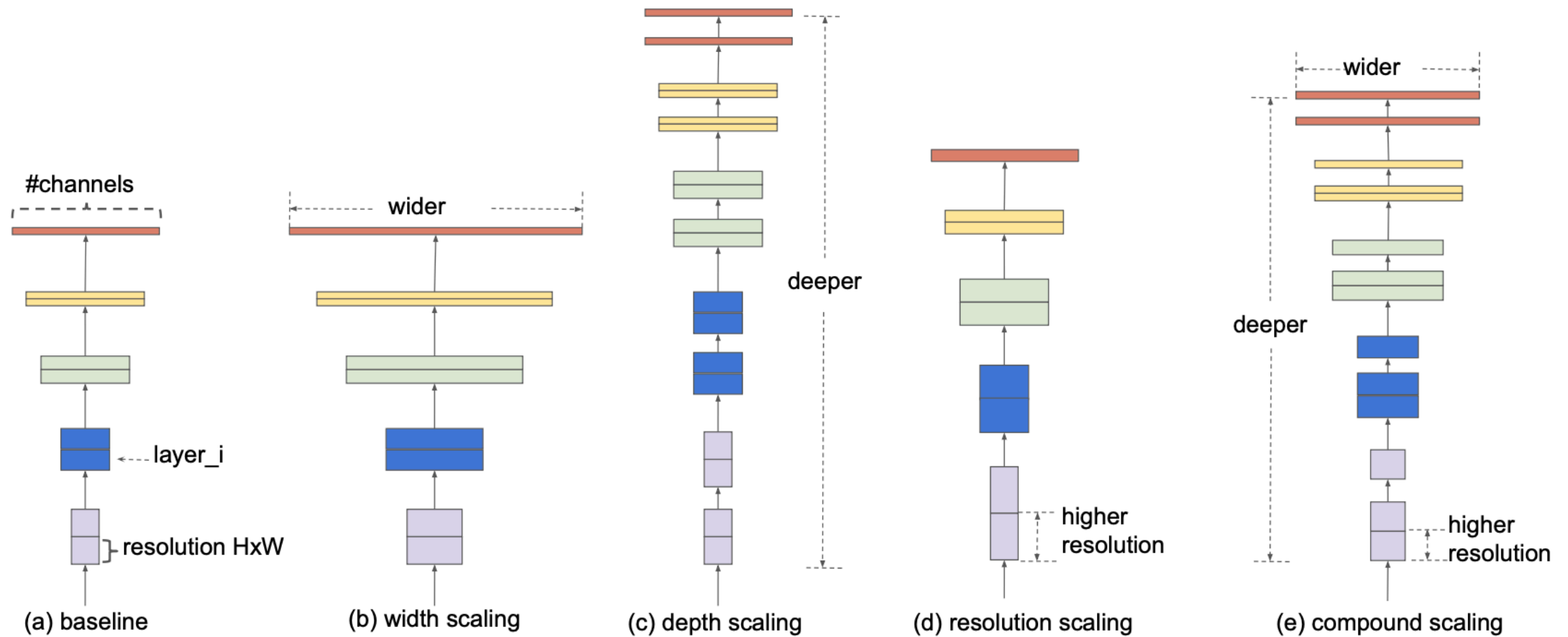


(b) Unraveled view of (a)

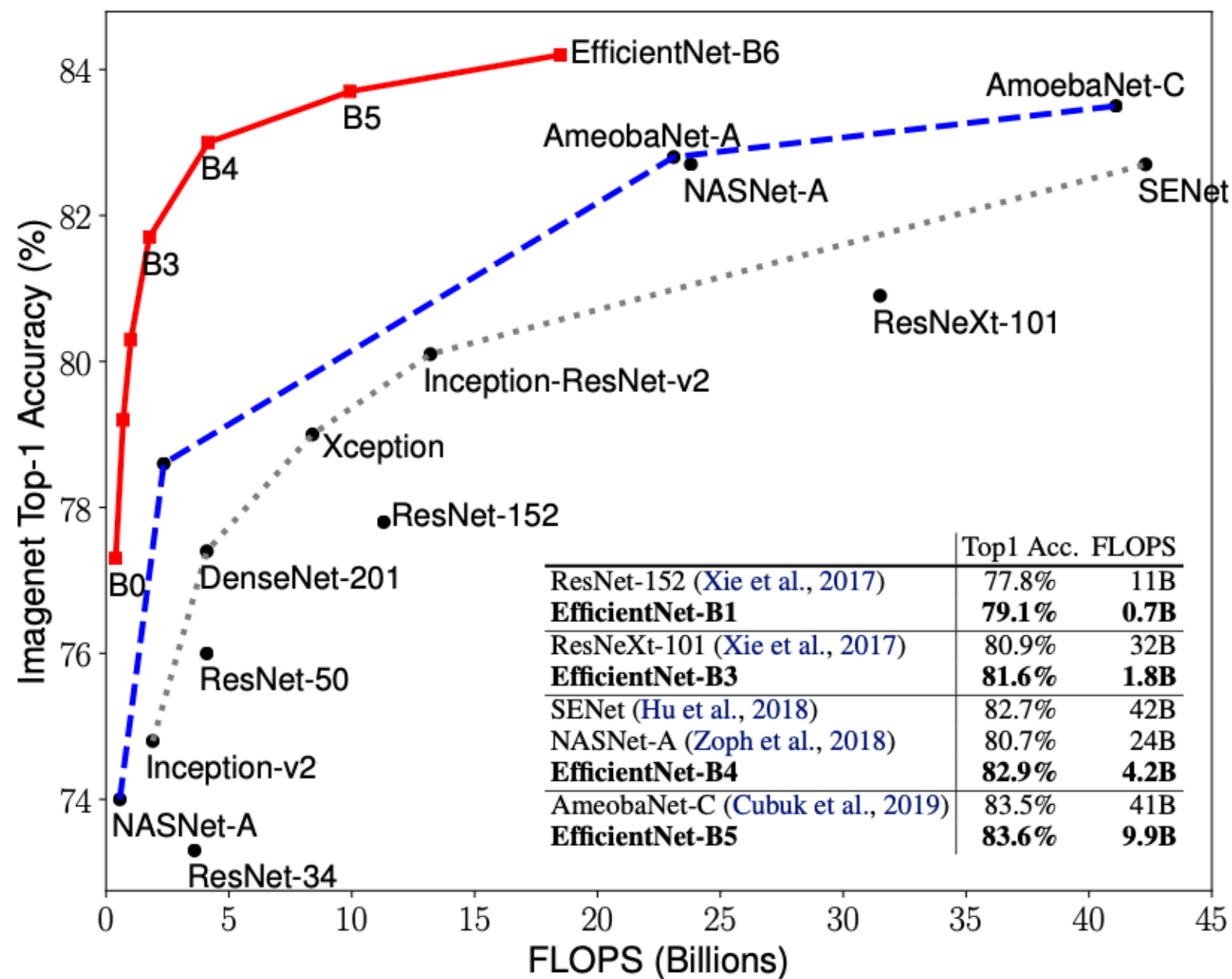
1. Veit, Wilber, & Belongie (2016 *NeurIPS*) ↩

EfficientNet

EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks¹



How good is EfficientNet?



Vision Transformer (ViT)

AN IMAGE IS WORTH 16X16 WORDS: TRANSFORMERS FOR IMAGE RECOGNITION AT SCALE

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^{*}equal technical contribution, [†]equal advising

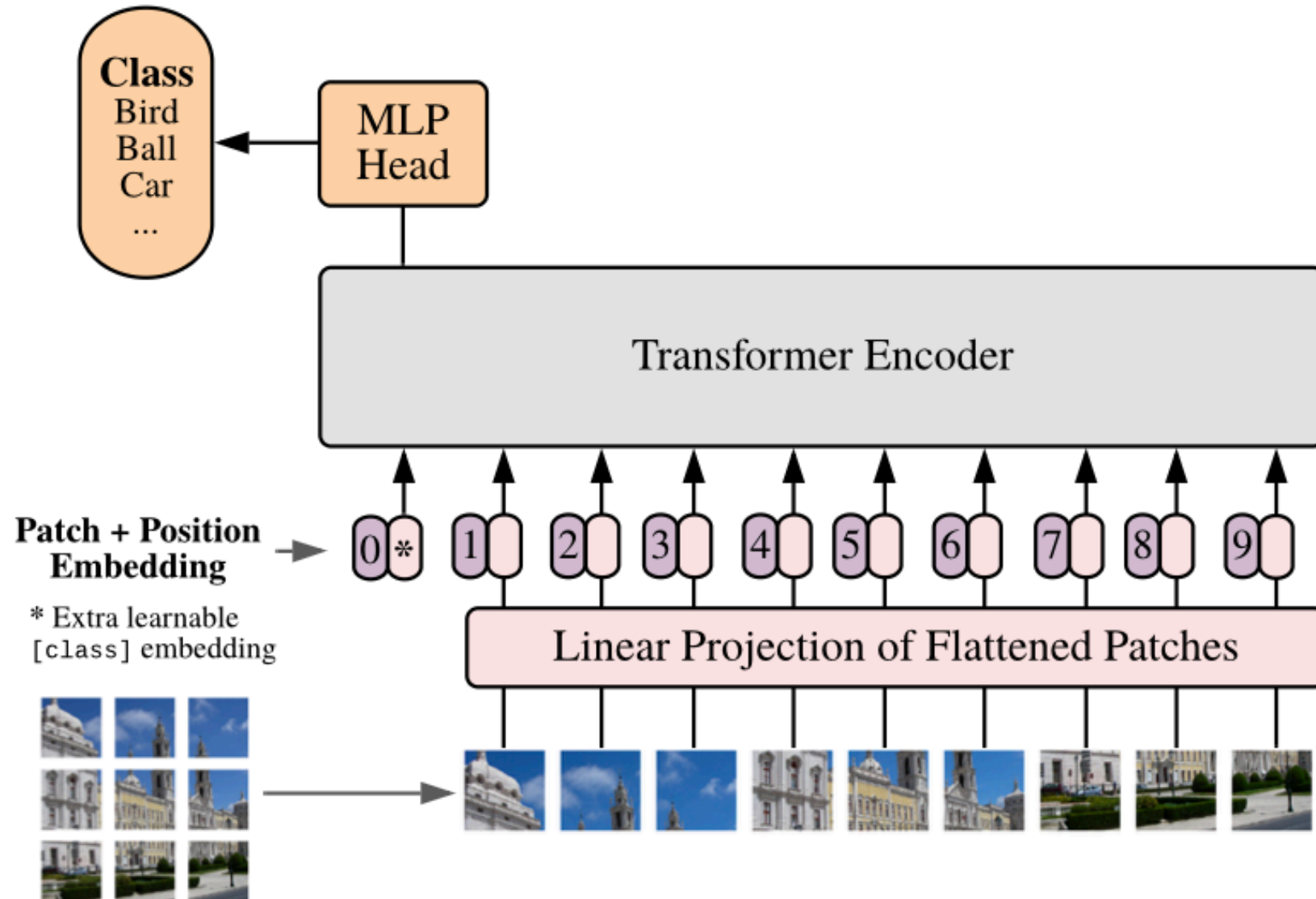
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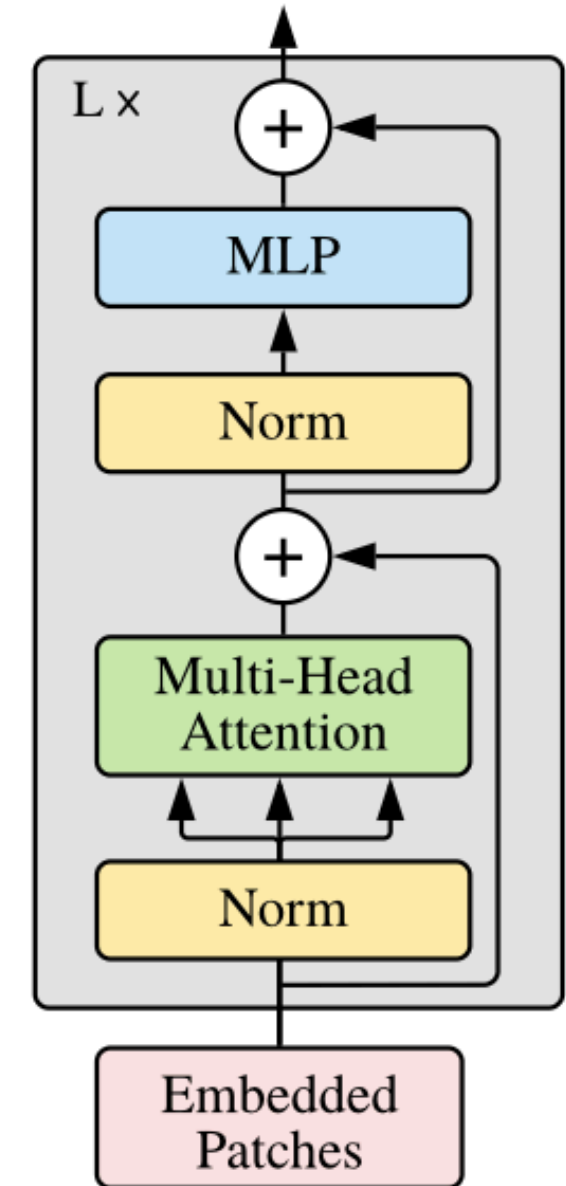
ABSTRACT

While the Transformer architecture has become the de-facto standard for natural language processing tasks, its applications to computer vision remain limited. In vision, attention is either applied in conjunction with convolutional networks, or used to replace certain components of convolutional networks while keeping their overall structure in place. We show that this reliance on CNNs is not necessary and a pure transformer applied directly to sequences of image patches can perform very well on image classification tasks. When pre-trained on large amounts of data and transferred to multiple mid-sized or small image recognition benchmarks (ImageNet, CIFAR-100, VTAB, etc.), Vision Transformer (ViT) attains excellent results compared to state-of-the-art convolutional networks while requiring substantially fewer computational resources to train.¹

Vision Transformer (ViT)



Transformer Encoder



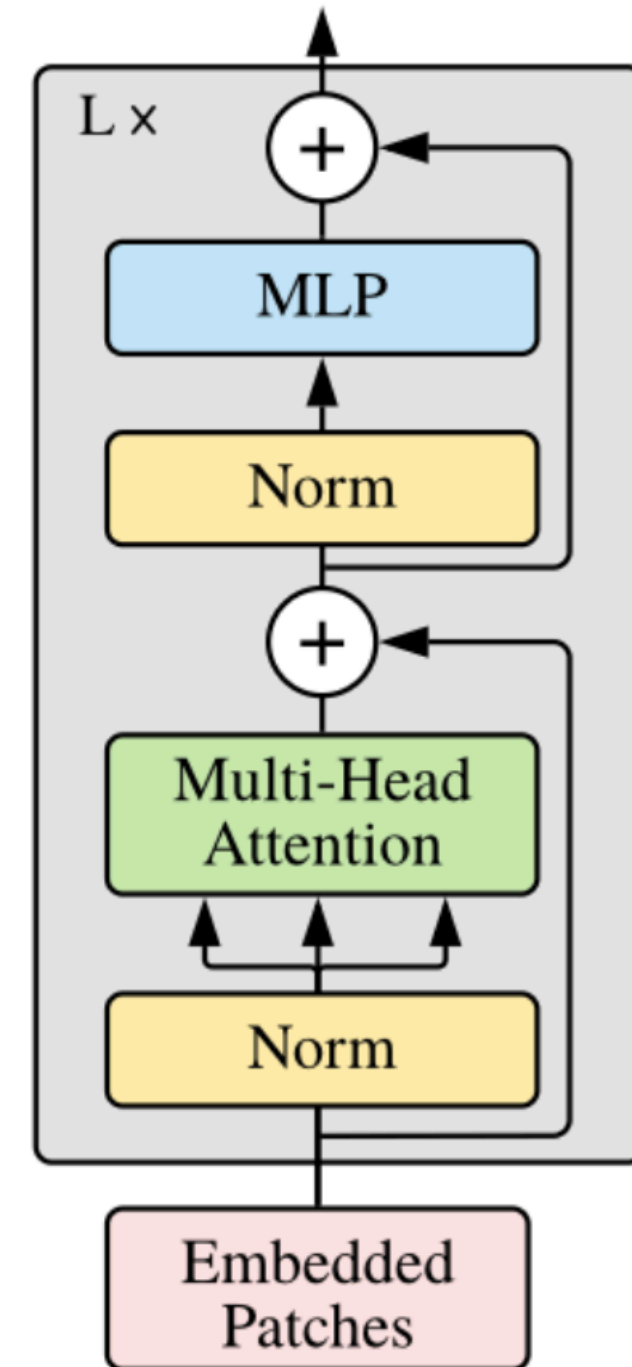
Transformer Encoder

composed of multi-head self
attention

and multi-layer perceptron modules

just the input are **embedded patches**

Transformer Encoder



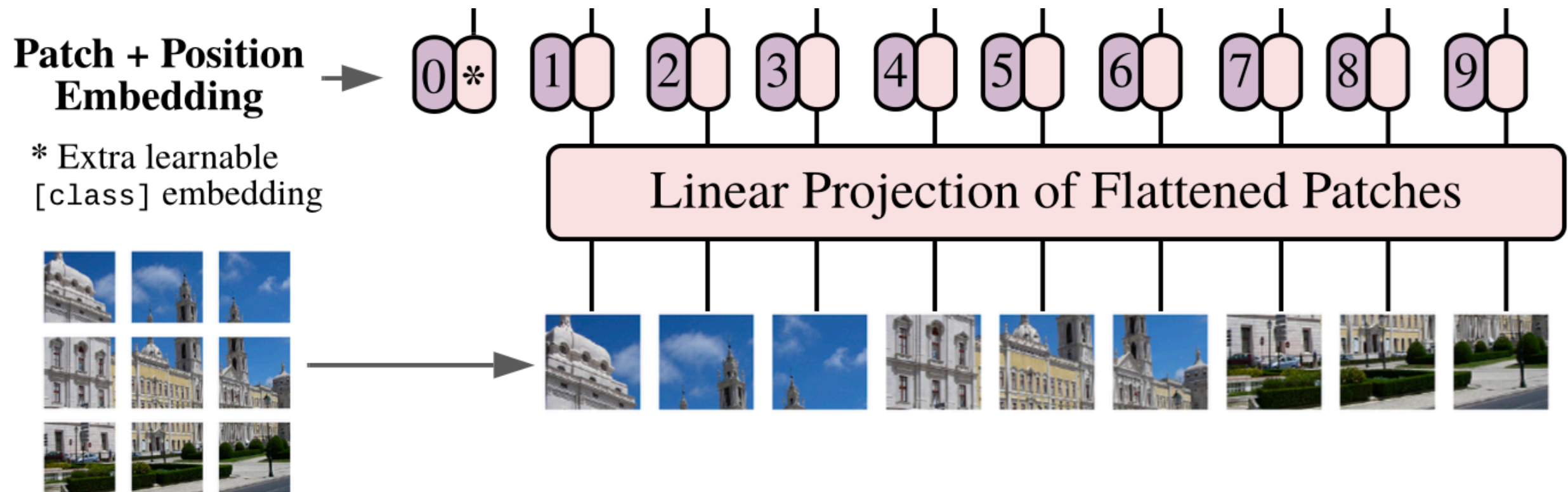
Embedded Patches

Split image into pre-defined number of patches

Apply a *learnable* linear projection

Add position embeddings, based on 2D position

Plus, add an special **classifier token**

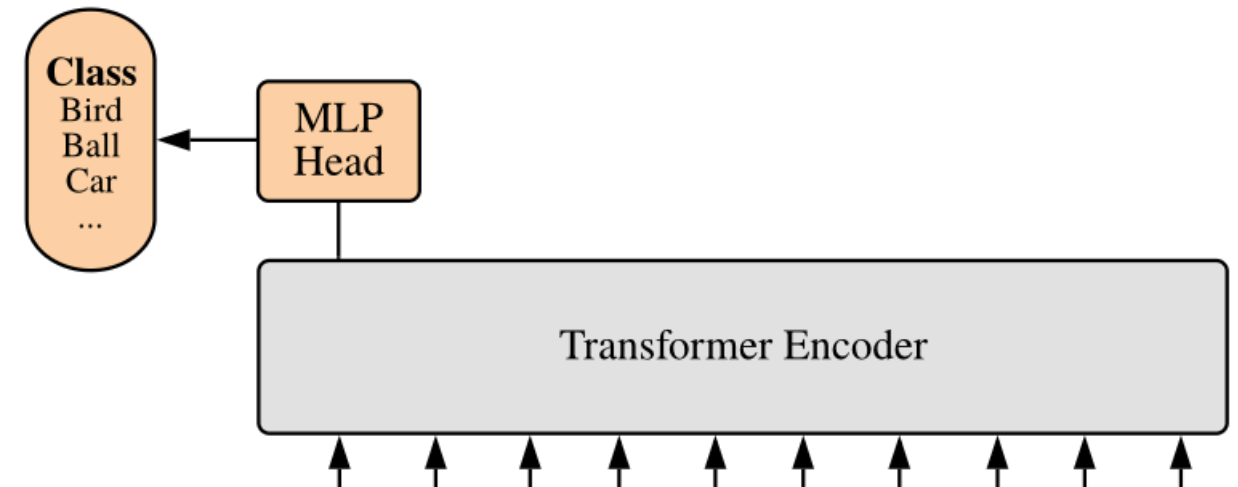


Classifier head

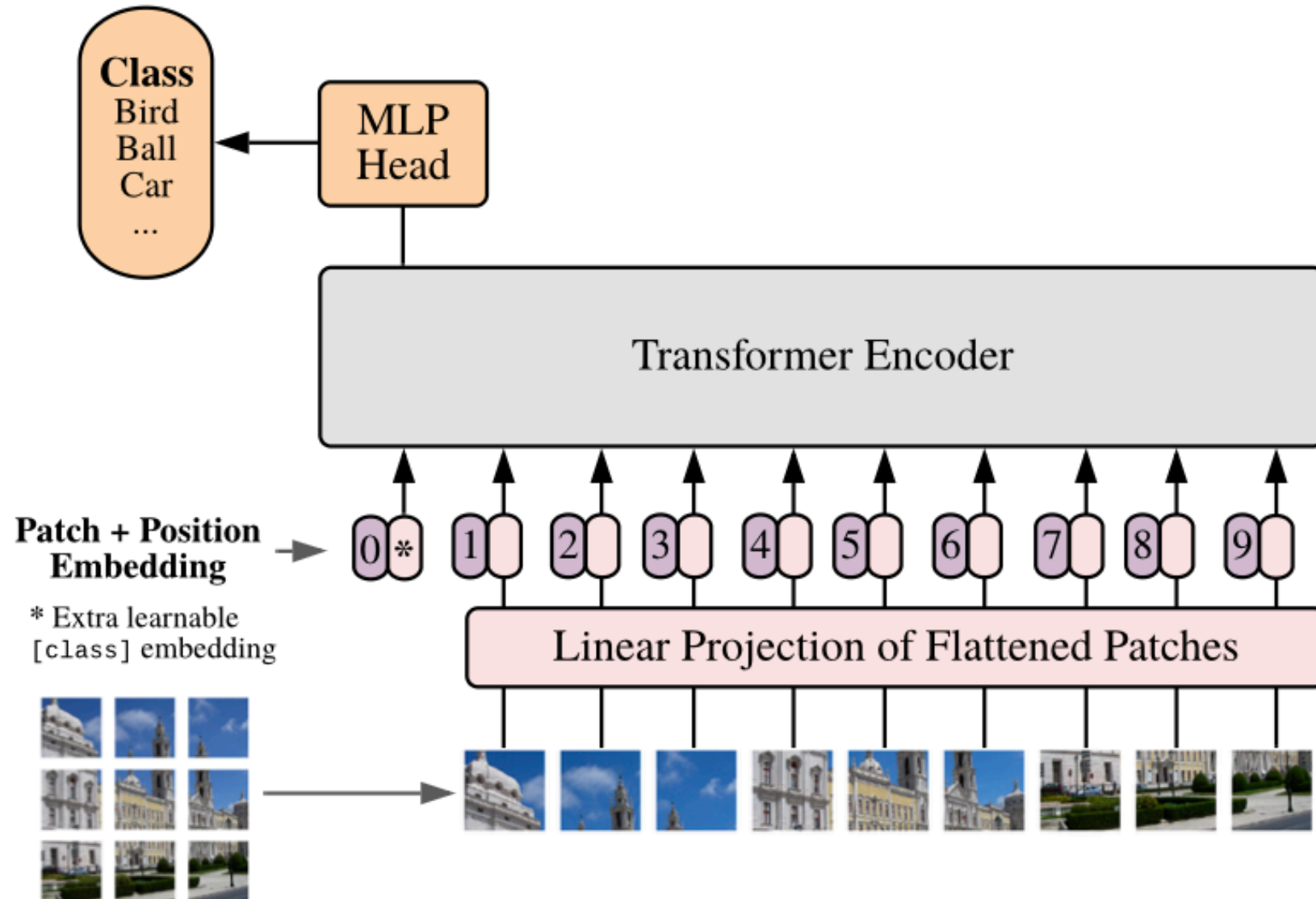
Here composed of a multi-layer perception

Attached on after all transformer layers

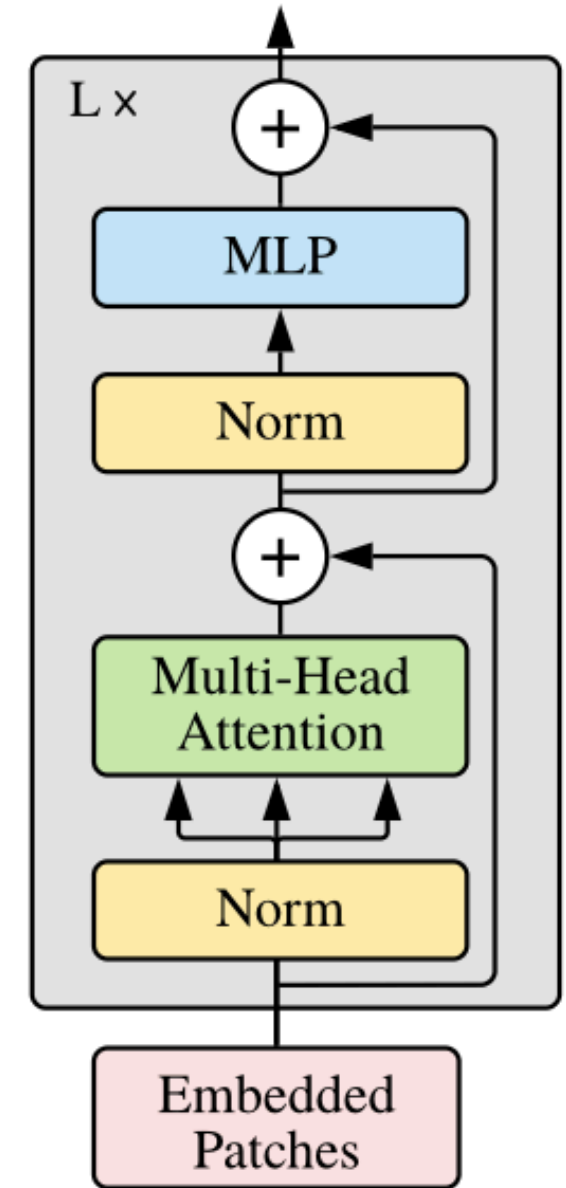
...at the position of the classifier token



Vision Transformer (ViT)



Transformer Encoder



ViT Training

Image classification on the ImageNet dataset.

Cross-entropy loss.

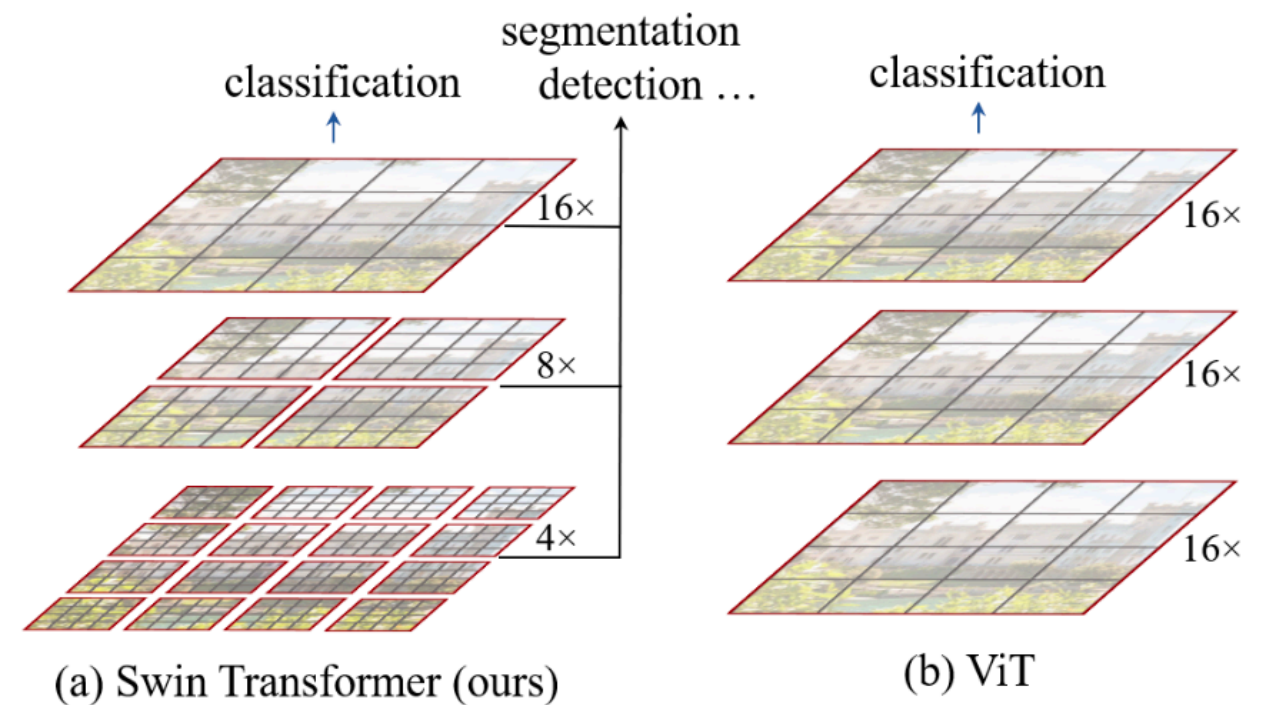
Swin Transformer

Swin Transformer

Hierarchical Vision Transformer using Shifted Windows¹

Linear complexity wrt input size

Self-attention only within each local window (red)



1. Liu et al. (2021) ↩

Layer 1



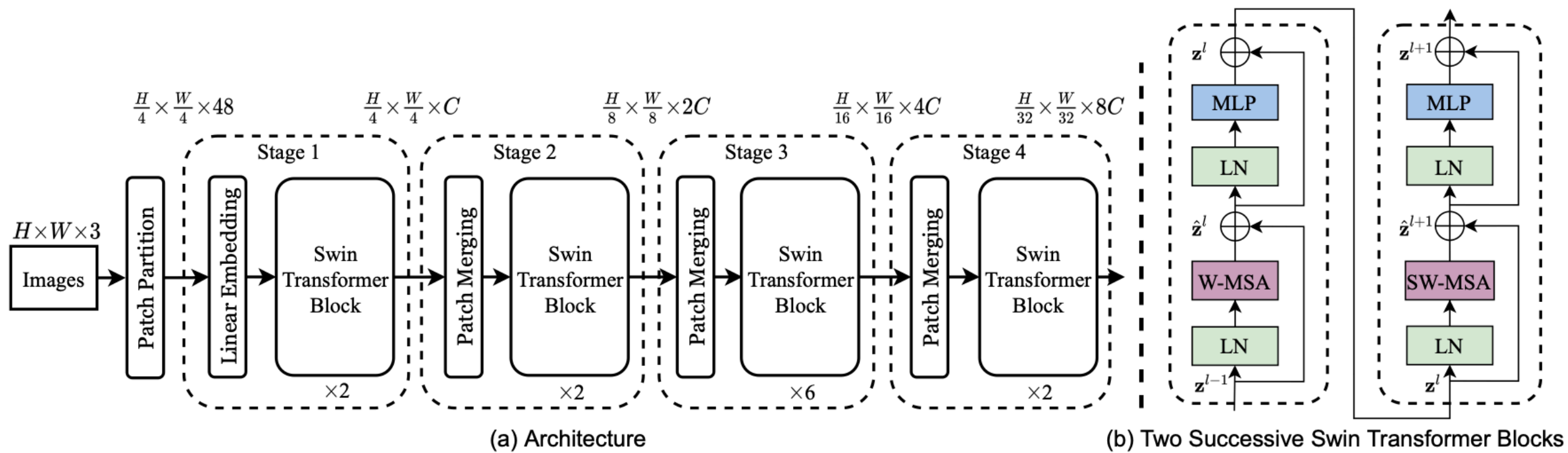
Layer 1+1



A local window to
perform self-attention

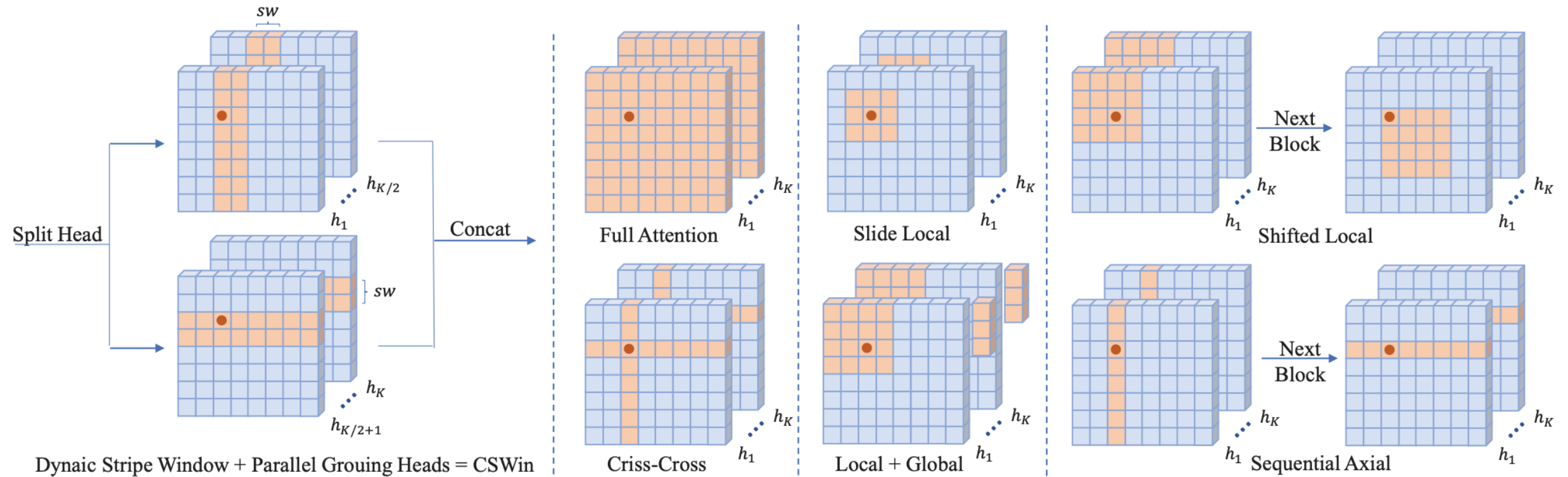


A patch

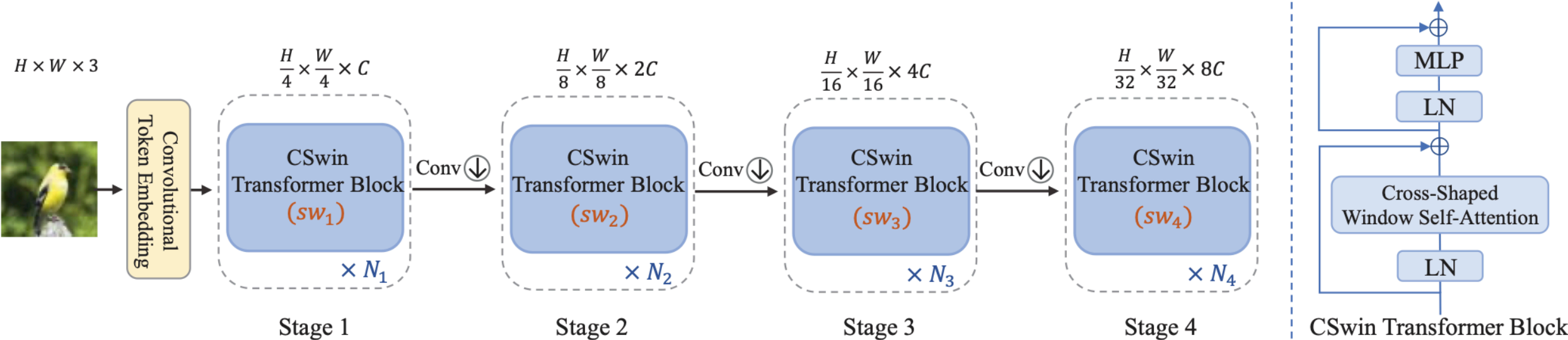


CSWin Transformer

CSWin¹: Cross-shaped window attention



CSWin: Architecture



CSWin vs Swin

Method	Image Size	#Param.	FLOPs	Throughput	Top-1
Eff-B4 [51]	380 ²	19M	4.2G	349/s	82.9
Eff-B5 [51]	456 ²	30M	9.9G	169/s	83.6
Eff-B6 [51]	528 ²	43M	19.0G	96/s	84.0
DeiT-S [53]	224 ²	22M	4.6G	940/s	79.8
DeiT-B [53]	224 ²	87M	17.5G	292/s	81.8
DeiT-B [53]	384 ²	86M	55.4G	85/s	83.1
PVT-S [58]	224 ²	25M	3.8G	820/s	79.8
PVT-M [58]	224 ²	44M	6.7G	526/s	81.2
PVT-L [58]	224 ²	61M	9.8G	367/s	81.7
T2T _t -14 [66]	224 ²	22M	6.1G	–	81.7
T2T _t -19 [66]	224 ²	39M	9.8G	–	82.2
T2T _t -24 [66]	224 ²	64M	15.0G	–	82.6
CvT-13 [60]	224 ²	20M	4.5G	–	81.6
CvT-21 [60]	224 ²	32M	7.1G	–	82.5
CvT-21 [60]	384 ²	32M	24.9G	–	83.3
Swin-T [38]	224 ²	29M	4.5G	755/s	81.3
Swin-S [38]	224 ²	50M	8.7G	437/s	83.0
Swin-B [38]	224 ²	88M	15.4G	278/s	83.3
Swin-B [38]	384 ²	88M	47.0G	85/s	84.2
CSWin-T	224 ²	23M	4.3G	701/s	82.7
CSWin-S	224 ²	35M	6.9G	437/s	83.6
CSWin-B	224 ²	78M	15.0G	250/s	84.2
CSWin-B	384 ²	78M	47.0G		85.4

Table 2. Comparison of different models on ImageNet-1K.

Representation Learning

What is representation learning?

In **Representation learning**, we focus on learning *good* representations of the data, regardless of what concrete tasks the model needs to solve.

As a result, a single model can be often be re-used for many different tasks.

Pre-trained image classifiers are excellent **backbones** for other architectures

1. Pre-train a model on a large image classification dataset (e.g., ImageNet)
2. Use these model to convert images into a vector representation (usually, the penultimate layer's representation)
3. Learn another machine learning model on top of these representations (image encoder can be **frozen** or **fine-tuned**)

Practical Things

Practical example: ViT classifier¹

```
from transformers import ViTImageProcessor, ViTForImageClassification
from PIL import Image
import requests

url = 'http://images.cocodataset.org/val2017/000000039769.jpg'
image = Image.open(requests.get(url, stream=True).raw)

processor = ViTImageProcessor.from_pretrained('google/vit-base-patch16-224')
model = ViTForImageClassification.from_pretrained('google/vit-base-patch16-224')

inputs = processor(images=image, return_tensors="pt")
outputs = model(**inputs)
logits = outputs.logits
# model predicts one of the 1000 ImageNet classes
predicted_class_idx = logits.argmax(-1).item()
print("Predicted class:", model.config.id2label[predicted_class_idx])
```

1. huggingface.co/google/vit-base-patch16-224 ↩

For representations, go with ViTModel instead¹

ViTModel

 `class transformers.ViTModel` 

[<source>](#)

```
( config: ViTConfig, add_pooling_layer: bool = True, use_mask_token: bool = False )
```

Parameters

- **config** ([ViTConfig](#)) — Model configuration class with all the parameters of the model. Initializing with a config file does not load the weights associated with the model, only the configuration. Check out the [from_pretrained\(\)](#) method to load the model weights.
- **add_pooling_layer** (`bool`, *optional*, defaults to `True`) — Whether to add a pooling layer
- **use_mask_token** (`bool`, *optional*, defaults to `False`) — Whether to use a mask token for masked image modeling.

The bare ViT Model outputting raw hidden-states without any specific head on top.

This model inherits from [PreTrainedModel](#). Check the superclass documentation for the generic methods the library implements for all its model (such as downloading or saving, resizing the input embeddings, pruning heads etc.)

This model is also a PyTorch [torch.nn.Module](#) subclass. Use it as a regular PyTorch Module and refer to the PyTorch documentation for all matter related to general usage and behavior.

1. https://huggingface.co/docs/transformers/en/model_doc/vit 

Let's have a look at the code

You can pre-train a ViT Model with different objectives

- **Image classification** (the classic)
- **Masked image modeling** (MIM; more like language modeling)
- **Self-distillation** (matching the output feature distribution of different augmented views of the same image)
- **Contrastive learning** (aligning image representations with text descriptions)

💡 *More later when we discuss Vision-Language Models* ➡ SOON

Summary

ViT is increasingly considered the state of the art in vision representations, but a bit more expensive than alternatives like EfficientNet

ViT can be pre-trained in various ways to learn good representations of images, though a model pre-trained on ImageNet is already fairly strong.

For downstream use, a common paradigm is to use a pre-trained model (backbone) and train a lightweight linear or MLP classifier on top of the learned representations. This paradigm is called **Transfer Learning**

Keywords

- Image classification
- Residual Networks (ResNet)
- EfficientNet
- Vision Transformer (ViT)
- Swin Transformer
- Representation learning

Resources

- For general deep learning background and also including Transformers: The New Bishop
- The annotated transformer
- Relevant videos by @3blue1brown, e.g. the one on attention in Transformers